

Army pool sensitivity — complete run matrix checklist

Purpose: Compare how pool membership rules, excluding zero–talent-board peers, and peer context metric on plots change sorting index (panel 5, H_sort) and promotion-rate porch (panel 9, HERO).

Where you work: AWS 520 root — the folder named `Network_1P_shell/` that contains:

- `big_dfs/`
- `520_pipeline_cox_working.ipynb`
- `pipeline_config.py`
- `talent/re_entry/` (plot scripts and manifest)
- `scripts/backup_rename_suffix.sh`

What you compare for Alex (assortativity priority):

- Panel 5 — read `H_sort` in the overlap plot title.
- Panel 9 — read `HERO` porch shape (promotion rate vs peer-context bins).
- Record values in the **results table** at the end of this document.

Vocabulary (read once)

Plain English	Setting or column name
Legacy pool grain	<code>POOL_GROUPING_MODE = 'legacy'</code> — peers share the same HRC snapshot date and the same senior-rater identifier (<code>snr_rater_bwd</code>).
Eval-thru pool grain	<code>POOL_GROUPING_MODE = 'active_at_eval_thru'</code> — peers are senior-rater officers whose OER evaluation window covers this row's OER completion date (<code>eval_thru_dt_bwd</code>).
Snapshot pool grain	<code>POOL_GROUPING_MODE = 'active_at_snapshot'</code> — peers are senior-rater officers whose OER window covers this row's HRC snapshot date (<code>snpsht_dt</code>).
Exact rating-window pool grain	<code>POOL_GROUPING_MODE = 'rating_window'</code> — peers share the same snapshot, senior rater, and identical backward eval window (<code>eval_strt/thru_bwd</code>). Often tiny or empty buckets; use as an empirical probe, not the main story.
Keep zero–TB peers in pool	<code>POOL_EXCLUDE_PEER_TB_ZERO = False</code>
Drop zero–TB peers from pool mean and size	<code>POOL_EXCLUDE_PEER_TB_ZERO = True</code> — removes peers with <code>tb_ratio == 0</code> from pool statistics; does not remove the rated officer from the cohort.
Straight leave-one-out pond level (T_j)	Feather column <code>pool_tb_ratio_mean_snr_fwd</code> — mean senior-rater pool talent, leave-one-out.
Minus-mean peer context	Feather column <code>pool_minus_mean_snr_fwd</code> — own talent minus pool mean.
Z-scored minus-mean (HERO default for minus-mean runs)	Feather column <code>z_pool_minus_mean_snr_fwd</code>
Live feather file plots read	<code>big_dfs/df_pipeline_11_cox_analysis.feather</code>
Data story mosaic output	<code>talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3.png</code>

Plot tag `_run1` in filenames: All five runs write PNGs with suffix `_run1` in the filename. That is a **plot tag**, not “Run 1 only.” Each sensitivity run **overwrites** the top-level PNGs until you **backup** into a subfolder.

Core matrix (Runs 1–5) — overview

Run label	Pool grain	Zero–TB peers	Peer metric on plots	Re-run notebook?
Run 1	legacy	keep	straight leave-one-out pond level	Yes
Run 2	eval-thru	keep	straight leave-one-out pond level	Yes
Run 3	eval-thru	exclude from pool	straight leave-one-out pond level	Yes
Run 4 (optional)	legacy (from archived feather)	keep	minus-mean	No — plots only
Run 5 (optional)	eval-thru (from archived feather)	keep	minus-mean	No — plots only

Optional exploration (Runs 6–8) — config only on old PDE / Army Virtual Desktop

No new Python on AWS: only `pipeline_config.py`, then the **520 notebook** + plot scripts (straight leave-one-out). **Each optional run below is a full cold-start checklist** — you can redo Run 6, 7, or 8 alone without opening another run section.

Run label	Pool grain	Zero-TB peers	What you learn
Run 6	snapshot	keep	Panel-date anchor vs Run 2 (eval-thru) — same SNR, different “who counts as peer.”
Run 7	rating_window	keep	Empirical check: how often exact shared windows yield usable pools (expect many tiny buckets).
Run 8	snapshot	exclude from pool	Mirror Run 3 at snapshot grain — is peer-TB-zero effect anchor-dependent ?

If time is tight: do Run 6 + Run 8; skip Run 7 unless you want a one-line pool-size fact for the memo.

Not on old PDE without Mac upload: own-officer TB≠0 plot cohort, weighted window pools, renamed config flags — see `BAND_OF_EXCELLENCE_ARMY_PORCH.md`.

Feather archives you must save after notebook runs:

Archive file	Create after
<code>big_dfs/df_pipeline_11_run1_legacy.feather</code>	Run 1 notebook
<code>big_dfs/df_pipeline_11_run2_thru.feather</code>	Run 2 notebook
<code>big_dfs/df_pipeline_11_run3_thru_nozero.feather</code>	Run 3 notebook
<code>big_dfs/df_pipeline_11_run6_activesnap.feather</code>	Run 6 notebook (optional)
<code>big_dfs/df_pipeline_11_run7_ratingwindow.feather</code>	Run 7 notebook (optional)
<code>big_dfs/df_pipeline_11_run8_activesnap_nozero.feather</code>	Run 8 notebook (optional)

Plot backup (`backup_rename_suffix.sh`) **does not** copy feathers. You must run the `cp` commands in each run’s save step.

Shared reference — plot script lines

These files live under `talent/re_entry/` on AWS.

Straight leave-one-out pond level (Runs 1, 2, 3, 6, 7, 8)

File	Line	Must read
<code>army_basic_plots.py</code>	42	<code>COL_L00 = "pool_tb_ratio_mean_snr_fwd"</code>
<code>army_hero_slide_plot.py</code>	44	<code>plot_var = "pool_tb_ratio_mean_snr_fwd"</code>
<code>army_act2_probes.py</code>	55	<code>COL_L00_Z = "z_pool_tb_ratio_mean_snr_fwd"</code> <i>(column usually missing → Act II z-scores <code>COL_L00</code> from basic plots)</i>

HERO command-line flag (Runs 1–3, 6–8):

```
python talent/re_entry/army_hero_slide_plot.py --plot-var pool_tb_ratio_mean_snr_fwd
```

Manifest panel 9 path (Runs 1–3, 6–8):

```
talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
```

Do not use `--plot-var pool_tb_ratio_mean_snr_fwd_run1.png` — that is a filename, not a column name.

Do not use `z_pool_tb_ratio_mean_snr_fwd` for Runs 1–3 or 6–8 unless you add that column via a separate Cell 10.5 config path.

Minus-mean peer context (Runs 4, 5)

File	Line	Must read
<code>army_basic_plots.py</code>	42	<code>COL_L00 = "pool_minus_mean_snr_fwd"</code>
<code>army_hero_slide_plot.py</code>	44	<code>plot_var = "z_pool_minus_mean_snr_fwd"</code>
<code>army_act2_probes.py</code>	55	<code>COL_L00_Z = "z_pool_minus_mean_snr_fwd"</code>

HERO command-line flag (Runs 4, 5):

```
python talent/re_entry/army_hero_slide_plot.py --plot-var z_pool_minus_mean_snr_fwd
```

Manifest panel 9 path (Runs 4, 5):

```
talent/re_entry/output/hero/ARMY_HERO_ew8_z_pool_minus_mean_snr_fwd_run1.png
```

Plot pipeline commands (every run that builds plots)

Run from `Network_1P_shell/`:

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var <COLUMN_NAME>
python talent/re_entry/build_army_data_story.py
```

Use `python`, not `bash`, for `build_army_data_story.py`.

Success lines:

- `Wrote talent/re_entry/output/hero/ARMY_HERO_ew8_...`
- `Wrote talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3.png`

Manifest file

Path: `talent/re_entry/manifests/army_run1_3x3_manifest.json`

Edit panel 9 `"path"` to match the HERO PNG filename **exactly** before running `build_army_data_story.py`.

Panels 2–8 paths in the manifest stay the same; scripts overwrite top-level PNGs.

Notebook cell order (every run that re-runs the notebook)

Open `520_pipeline_cox_working.ipynb`. Run in order:

1. **Cell 0** — reloads `pipeline_config.py`
2. **Cell 5** — pool means (watch log for `POOL_GROUPING_MODE`)
3. **Cell 6** — or if Cell 6 fails, run in terminal:

```
cp big-dfs/df_pipeline_05_pool_means.feather big-dfs/df_pipeline_06_pool_ranks.feather
```

4. **Cell 7**
5. **Cell 8**
6. **Cell 9**
7. **Cell 10**
8. **Cell 10.5**
9. **Cell 11** — writes `big-dfs/df_pipeline_11_cox_analysis.feather`

Feather column check (after Cell 11)

```
python3 -c "import pandas as pd; df=pd.read_feather('big-dfs/df_pipeline_11_cox_analysis.feather');
print('pool_tb_ratio_mean_snr_fwd', 'YES' if 'pool_tb_ratio_mean_snr_fwd' in df.columns else 'MISSING')"
```

For Runs 4 and 5 also check:

```
python3 -c "import pandas as pd; df=pd.read_feather('big-dfs/df_pipeline_11_cox_analysis.feather');
print('z_pool_minus_mean_snr_fwd', 'YES' if 'z_pool_minus_mean_snr_fwd' in df.columns else 'MISSING')"
```

RUN 1 • Legacy pool grain • straight leave-one-out • keep zero–TB peers

What this run isolates: Baseline legacy pool definition with **pond-level** peer context on plots.

Run 1 — Step 1 • Open terminal on AWS

```
cd /path/to/Network_1P_shell
conda activate TALNET39
date
ls -lh big-dfs/df_pipeline_11_cox_analysis.feather
```

The last line is optional; it shows whether an old feather exists before you rebuild.

Run 1 — Step 2 • Edit `pipeline_config.py`

Open `pipeline_config.py` at 520 root. Set these lines **exactly**:

```
POOL_GROUPING_MODE = 'legacy'
POOL_EXCLUDE_PEER_TB_ZERO = False
CELL5_POOL_MEANS = True
CELL6_POOL_RANKS = True
```

Save the file.

Also confirm these lines exist (needed by Cell 5; add if missing):

```
POOL_ANCHOR_COL = 'eval_thru_dt_bwd'
POOL_EVAL_STRT_COL = 'eval_strt_dt_bwd'
POOL_EVAL_THRU_COL = 'eval_thru_dt_bwd'
```

Run 1 — Step 3 · Edit plot scripts (straight leave-one-out)

File: `talent/re_entry/army_basic_plots.py` — line 42:

```
COL_L00 = "pool_tb_ratio_mean_snr_fwd"
```

File: `talent/re_entry/army_hero_slide_plot.py` — line 44:

```
plot_var = "pool_tb_ratio_mean_snr_fwd"
```

File: `talent/re_entry/army_act2_probes.py` — line 55 (optional but recommended for consistent panels 7–8):

```
COL_L00_Z = "z_pool_tb_ratio_mean_snr_fwd"
```

Save all edited files.

Run 1 — Step 4 · Run notebook cells

Open `520_pipeline_cox_working.ipynb`.

Run **Cell 0**, then **Cell 5**, **Cell 6** (or the `cp` fallback), **Cell 7**, **Cell 8**, **Cell 9**, **Cell 10**, **Cell 10.5**, **Cell 11**.

In the Cell 5 log, confirm:

```
POOL_GROUPING_MODE='legacy'
```

Run 1 — Step 5 · Check feather columns

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');
print('pool_tb_ratio_mean_snr_fwd', 'YES' if 'pool_tb_ratio_mean_snr_fwd' in df.columns else 'MISSING')"
```

Expect: `YES`.

Run 1 — Step 6 · Edit manifest panel 9

Open `talent/re_entry/manifests/army_run1_3x3_manifest.json`.

Set panel 9 `"path"` to:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
```

Save the manifest.

Run 1 — Step 7 · Run plot scripts and mosaic

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var pool_tb_ratio_mean_snr_fwd
python talent/re_entry/build_army_data_story.py
```

Run 1 — Step 8 · Verify Run 1

```
grep POOL_GROUPING_MODE pipeline_config.py
ls -l talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
ls -l talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3.png
```

In the basic plots log, find:

```
Overlap pool grain: POOL_GROUPING_MODE='legacy'
```

Open panel 5 — note **H_{sort}**. Open panel 9 — note porch shape.

Compare mosaic to standalone PNGs at **top level** under `talent/re_entry/output/` (not backup subfolders).

Run 1 — Step 9 · Archive feather and plots before Run 2

Save feather (required for Run 4 later):

```
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run1_legacy.feather
```

Save plots:

```
bash scripts/backup_rename_suffix.sh _run1_legacy --all-output
```

RUN 2 · Eval-thru pool grain · straight leave-one-out · keep zero–TB peers

What this run isolates: Same plot peer metric as Run 1, but eval-thru pool grain — tests assortativity / H_sort effect of pool definition.

Run 2 — Step 1 · Open terminal on AWS

```
cd /path/to/Network_1P_shell
conda activate TALNET39
date
```

Run 2 — Step 2 · Edit `pipeline_config.py`

```
POOL_GROUPING_MODE = 'active_at_eval_thru'
POOL_EXCLUDE_PEER_TB_ZERO = False
CELL5_POOL_MEANS = True
CELL6_POOL_RANKS = True
```

Save the file.

Run 2 — Step 3 · Edit plot scripts (straight leave-one-out)

File: `talent/re_entry/army_basic_plots.py` — line 42:

```
COL_L00 = "pool_tb_ratio_mean_snr_fwd"
```

File: `talent/re_entry/army_hero_slide_plot.py` — line 44:

```
plot_var = "pool_tb_ratio_mean_snr_fwd"
```

File: `talent/re_entry/army_act2_probes.py` — line 55:

```
COL_L00_Z = "z_pool_tb_ratio_mean_snr_fwd"
```

Save all edited files.

Run 2 — Step 4 · Run notebook cells

Open `520_pipeline_cox_working.ipynb`.

Run Cell 0, Cell 5, Cell 6 (or `cp` fallback), Cell 7, Cell 8, Cell 9, Cell 10, Cell 10.5, Cell 11.

In the Cell 5 log, confirm:

```
POOL_GROUPING_MODE='active_at_eval_thru'
```

Run 2 — Step 5 · Check feather columns

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');
print('pool_tb_ratio_mean_snr_fwd', 'YES' if 'pool_tb_ratio_mean_snr_fwd' in df.columns else 'MISSING')"
```

Expect: `YES`.

Run 2 — Step 6 · Edit manifest panel 9

Open `talent/re_entry/manifests/army_run1_3x3_manifest.json`.

Set panel 9 `"path"` to:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
```

Save the manifest.

Run 2 — Step 7 · Run plot scripts and mosaic

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var pool_tb_ratio_mean_snr_fwd
python talent/re_entry/build_army_data_story.py
```

Run 2 — Step 8 · Verify Run 2

```
grep POOL_GROUPING_MODE pipeline_config.py
ls -l talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
ls -l talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3.png
```

In the basic plots log, find:

```
Overlap pool grain: POOL_GROUPING_MODE='active_at_eval_thru'
```

Compare panel 5 `H_sort` and panel 9 porch to Run 1 backup in `talent/re_entry/output/*/run1_legacy/`.

Run 2 — Step 9 · Archive feather and plots before Run 3

Save feather (required for Run 5 later):

```
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run2_thru.feather
```

Save plots:

```
bash scripts/backup_rename_suffix.sh _run2_thru --all-output
```

RUN 3 · Eval-thru pool grain · straight leave-one-out · exclude zero–TB peers

What this run isolates: Same eval-thru grain as Run 2, but peers with talent board ratio equal to zero are dropped from pool mean and pool size.

Run 3 — Step 1 · Open terminal on AWS

```
cd /path/to/Network_1P_shell
conda activate TALNET39
date
```

Run 3 — Step 2 · Edit `pipeline_config.py`

```
POOL_GROUPING_MODE = 'active_at_eval_thru'
POOL_EXCLUDE_PEER_TB_ZERO = True
CELL5_POOL_MEANS = True
CELL6_POOL_RANKS = True
```

Save the file.

Run 3 — Step 3 · Edit plot scripts (straight leave-one-out)

File: `talent/re_entry/army_basic_plots.py` — line 42:

```
COL_L00 = "pool_tb_ratio_mean_snr_fwd"
```

File: talent/re_entry/army_hero_slide_plot.py — line 44:

```
plot_var = "pool_tb_ratio_mean_snr_fwd"
```

File: talent/re_entry/army_act2_probes.py — line 55:

```
COL_L00_Z = "z_pool_tb_ratio_mean_snr_fwd"
```

Save all edited files.

Run 3 — Step 4 · Run notebook cells

Open 520_pipeline_cox_working.ipynb .
Run Cell 0, Cell 5, Cell 6 (or cp fallback), Cell 7, Cell 8, Cell 9, Cell 10, Cell 10.5, Cell 11.
In the Cell 5 log, confirm:

```
POOL_GROUPING_MODE='active_at_eval_thru'
```

and, if printed:

```
POOL_EXCLUDE_PEER_TB_ZERO=True
```

Run 3 — Step 5 · Check feather columns

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');  
print('pool_tb_ratio_mean_snr_fwd', 'YES' if 'pool_tb_ratio_mean_snr_fwd' in df.columns else 'MISSING')"
```

Expect: YES .
Optional pool-size diagnostic:

```
python -u talent/re_entry/army_pool_size_probe.py
```

Run 3 — Step 6 · Edit manifest panel 9

Open talent/re_entry/manifests/army_run1_3x3_manifest.json .
Set panel 9 "path" to:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
```

Save the manifest.

Run 3 — Step 7 · Run plot scripts and mosaic

```
python talent/re_entry/army_basic_plots.py --all  
python talent/re_entry/army_act2_probes.py --plot all_probes  
python talent/re_entry/army_hero_slide_plot.py --plot-var pool_tb_ratio_mean_snr_fwd  
python talent/re_entry/build_army_data_story.py
```

Run 3 — Step 8 · Verify Run 3

Panel 5 title should include exclude_tb_ratio=0 peers .
Compare panel 5 H_sort and panel 9 porch to Run 2 backup in talent/re_entry/output/*/run2_thru/ .
Panel 2 left histogram may still show a spike at own talent equal to zero — that is own performance, not the pool-exclusion toggle.

Run 3 — Step 9 · Archive feather and plots

Save feather:

```
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run3_thru_nozero.feather
```

Save plots:

```
bash scripts/backup_rename_suffix.sh _run3_thru_nozero --all-output
```

RUN 4 · Legacy pool grain · minus-mean plots · plot-only (no notebook)

What this run isolates: Minus-mean peer context on plots, using the same feather as Run 1 (legacy grain). Do not re-run the notebook unless the Run 1 archive is missing.

Run 4 — Step 1 · Open terminal on AWS

```
cd /path/to/Network_1P_shell
conda activate TALNET39
date
ls -lh big_dfs/df_pipeline_11_run1_legacy.feather
```

If the archive file does not exist, stop and complete Run 1 through Run 1 Step 9 first.

Run 4 — Step 2 · Restore Run 1 feather into the live slot

Plot scripts always read big_dfs/df_pipeline_11_cox_analysis.feather . After Run 3, that file holds Run 3 data. Copy the Run 1 archive back:

```
cp big_dfs/df_pipeline_11_run1_legacy.feather big_dfs/df_pipeline_11_cox_analysis.feather
```

Direction: run1_legacy → cox_analysis (restore, not save).

Run 4 — Step 3 · pipeline_config.py

No edit required for plot-only Run 4 — pool columns are already in the restored feather.

Run 4 — Step 4 · Edit plot scripts (minus-mean)

File: talent/re_entry/army_basic_plots.py — line 42:

```
COL_L00 = "pool_minus_mean_snr_fwd"
```

File: talent/re_entry/army_hero_slide_plot.py — line 44:

```
plot_var = "z_pool_minus_mean_snr_fwd"
```

File: talent/re_entry/army_act2_probes.py — line 55:

```
COL_L00_Z = "z_pool_minus_mean_snr_fwd"
```

Save all edited files.

Run 4 — Step 5 · Do not run the notebook

Skip Cells 0 through 11.

Run 4 — Step 6 · Check feather columns

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');
print('z_pool_minus_mean_snr_fwd', 'YES' if 'z_pool_minus_mean_snr_fwd' in df.columns else 'MISSING')"
```

Expect: YES . If MISSING, the Run 1 notebook did not reach Cell 10.5 — re-run Run 1 notebook, then repeat Run 1 archive and restore steps.

Run 4 — Step 7 · Edit manifest panel 9

Open talent/re_entry/manifests/army_run1_3x3_manifest.json .

Set panel 9 "path" to:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_z_pool_minus_mean_snr_fwd_run1.png
```

Save the manifest.

Run 4 — Step 8 · Run plot scripts and mosaic

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var z_pool_minus_mean_snr_fwd
python talent/re_entry/build_army_data_story.py
```

Run 4 — Step 9 · Verify Run 4

Panel 5 **H_sort** should **match Run 1** (same legacy grain in feather — only panels 3, 7, 8, 9 peer axes change).
Panel 9 porch may differ from Run 1 straight leave-one-out HERO.

Run 4 — Step 10 · Archive plots

```
bash scripts/backup_rename_suffix.sh _run4_legacy_minusmean --all-output
```

RUN 5 · Eval-thru pool grain · minus-mean plots · plot-only (no notebook)

What this run isolates: Minus-mean peer context on plots, using the **same feather as Run 2** (eval-thru grain, keep zero–TB peers). **Do not re-run the notebook** unless the Run 2 archive is missing.

Run 5 — Step 1 · Open terminal on AWS

```
cd /path/to/Network_1P_shell
conda activate TALNET39
date
ls -lh big-dfs/df_pipeline_11_run2_thru.feather
```

If the archive file **does not exist**:

- 1. Re-run **Run 2** notebook (Run 2 Steps 2–4), then
- 2. Run **Run 2 Step 9** feather save:

```
cp big-dfs/df_pipeline_11_cox_analysis.feather big-dfs/df_pipeline_11_run2_thru.feather
```

Run 5 — Step 2 · Restore Run 2 feather into the live slot

```
cp big-dfs/df_pipeline_11_run2_thru.feather big-dfs/df_pipeline_11_cox_analysis.feather
```

Direction: `run2_thru` → `cox_analysis` (restore, not save).

Run 5 — Step 3 · `pipeline_config.py`

No edit required for plot-only Run 5 — pool columns are already in the restored feather.
For consistent panel 5 **labels** only (optional, does not change feather):

```
POOL_GROUPING_MODE = 'active_at_eval_thru'
POOL_EXCLUDE_PEER_TB_ZERO = False
```

Run 5 — Step 4 · Edit plot scripts (minus-mean)

File: `talent/re_entry/army_basic_plots.py` — line 42:

```
COL_L00 = "pool_minus_mean_snr_fwd"
```

File: `talent/re_entry/army_hero_slide_plot.py` — line 44:

```
plot_var = "z_pool_minus_mean_snr_fwd"
```

File: `talent/re_entry/army_act2_probes.py` — line 55:

```
COL_L00_Z = "z_pool_minus_mean_snr_fwd"
```

Save all edited files.

Run 5 — Step 5 · Do not run the notebook

Skip Cells 0 through 11.

Run 5 — Step 6 · Check feather columns

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');
print('z_pool_minus_mean_snr_fwd', 'YES' if 'z_pool_minus_mean_snr_fwd' in df.columns else 'MISSING')"
```

Expect: **YES**.

Run 5 — Step 7 · Edit manifest panel 9

Open `talent/re_entry/manifests/army_run1_3x3_manifest.json`.

Set panel 9 `"path"` to:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_z_pool_minus_mean_snr_fwd_run1.png
```

Save the manifest.

Run 5 — Step 8 · Run plot scripts and mosaic

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var z_pool_minus_mean_snr_fwd
python talent/re_entry/build_army_data_story.py
```

Run 5 — Step 9 · Verify Run 5

Panel 5 **H_sort** should **match Run 2** (same eval-thru grain in feather).
Compare panel 9 to Run 4 — minus-mean at **eval-thru** grain vs minus-mean at **legacy** grain.
Compare panel 9 to Run 2 — minus-mean vs straight leave-one-out at **eval-thru** grain.

Run 5 — Step 10 · Archive plots

```
bash scripts/backup_rename_suffix.sh _run5_thru_minusmean --all-output
```

RUN 6 · Snapshot pool grain · straight leave-one-out · keep zero–TB peers

What this run isolates: Straight leave-one-out pond level on plots; peers defined by `active_at_snapshot` (OER window covers `snpsht_dt`) — compare panel 5 **H_sort** to eval-thru grain when you have Run 2 archived.

Run 6 — Step 1 · Open terminal on AWS

```
cd /path/to/Network_1P_shell
conda activate TALNET39
date
```

Run 6 — Step 2 · Edit `pipeline_config.py`

```
POOL_GROUPING_MODE = 'active_at_snapshot'
POOL_EXCLUDE_PEER_TB_ZERO = False
CELL5_POOL_MEANS = True
CELL6_POOL_RANKS = True
```

Save the file.

Run 6 — Step 3 · Edit plot scripts (straight leave-one-out)

File: `talent/re_entry/army_basic_plots.py` — line 42:

```
COL_L00 = "pool_tb_ratio_mean_snr_fwd"
```

File: `talent/re_entry/army_hero_slide_plot.py` — line 44:

```
plot_var = "pool_tb_ratio_mean_snr_fwd"
```

File: `talent/re_entry/army_act2_probes.py` — line 55:

```
COL_L00_Z = "z_pool_tb_ratio_mean_snr_fwd"
```

Save all edited files.

Run 6 — Step 3b · One-time fix on old PDE (Cell 5 `KeyError: snpsht_dt`)

If Cell 5 crashes with `KeyError: 'snpsht_dt'` inside `_add_pool_mean_size_active_at_anchor`, the AWS copy of `add_cum_oer_metrics_mod_working.py` predates the snapshot merge fix. Edit that file at **520 root** (same folder as the notebook). Find the block that builds `peer_g` before `cross = left.merge(peer_g, on=rater_col, ...)`. Replace the old `peer_g = peers.rename(columns={...})` with:

```
peer_rename = {
    pid_col: '_peer_pid',
    eval_strt_col: '_peer_strt',
    eval_thru_col: '_peer_thru',
    value_col: '_peer_val',
}
if anchor_col not in (eval_strt_col, eval_thru_col):
    peer_rename[anchor_col] = '_peer_anchor'
peer_g = peers.rename(columns=peer_rename)
cross = left.merge(peer_g, on=rater_col, how='inner')
```

Save the file. Run **Cell 0** (reload modules), then continue at Step 4. **Run 8** uses the same mode — this patch covers both.

Run 6 — Step 4 · Run notebook cells

Open `520_pipeline_cox_working.ipynb`.

Run **Cell 0**, **Cell 5**, **Cell 6** (or `cp` fallback), **Cell 7**, **Cell 8**, **Cell 9**, **Cell 10**, **Cell 10.5**, **Cell 11**.

In the Cell 5 log, confirm:

```
POOL_GROUPING_MODE='active_at_snapshot'
```

Run 6 — Step 5 · Check feather columns

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');
print('pool_tb_ratio_mean_snr_fwd', 'YES' if 'pool_tb_ratio_mean_snr_fwd' in df.columns else 'MISSING')"
```

Expect: `YES`.

Run 6 — Step 6 · Edit manifest panel 9

Open `talent/re_entry/manifests/army_run1_3x3_manifest.json`.

Set panel 9 `"path"` to:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
```

Save the manifest.

Run 6 — Step 7 · Run plot scripts and mosaic

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var pool_tb_ratio_mean_snr_fwd
python talent/re_entry/build_army_data_story.py
```

Run 6 — Step 8 · Verify Run 6

```
grep POOL_GROUPING_MODE pipeline_config.py
grep POOL_EXCLUDE_PEER_TB_ZERO pipeline_config.py
ls -l talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
ls -l talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3.png
```

In the basic plots log, find:

```
Overlap pool grain: POOL_GROUPING_MODE='active_at_snapshot'
```

Record panel 5 **H_sort** and panel 9 porch notes. If you have Run 2 plots archived under `talent/re_entry/output/*/run2_thru/`, compare **H_sort** (snapshot anchor vs eval-thru anchor).

Run 6 — Step 9 · Archive feather and plots

Save feather:

```
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run6_activesnap.feather
```

Save plots:

```
bash scripts/backup_rename_suffix.sh _run6_activesnap --all-output
```

RUN 7 · Rating-window pool grain · straight leave-one-out · keep zero–TB peers

What this run isolates: Strictest `rating_window` pool grain already in config — documents how often exact shared eval windows yield usable pools (often tiny buckets).

Run 7 — Step 1 · Open terminal on AWS

```
cd /path/to/Network_1P_shell
conda activate TALNET39
date
```

Run 7 — Step 2 · Edit `pipeline_config.py`

```
POOL_GROUPING_MODE = 'rating_window'
POOL_EXCLUDE_PEER_TB_ZERO = False
CELL5_POOL_MEANS = True
CELL6_POOL_RANKS = True
```

Save the file.

Run 7 — Step 3 · Edit plot scripts (straight leave-one-out)

File: `talent/re_entry/army_basic_plots.py` — line 42:

```
COL_L00 = "pool_tb_ratio_mean_snr_fwd"
```

File: `talent/re_entry/army_hero_slide_plot.py` — line 44:

```
plot_var = "pool_tb_ratio_mean_snr_fwd"
```

File: `talent/re_entry/army_act2_probes.py` — line 55:

```
COL_L00_Z = "z_pool_tb_ratio_mean_snr_fwd"
```

Save all edited files.

Run 7 — Step 4 · Run notebook cells

Open `520_pipeline_cox_working.ipynb`.

Run Cell 0, Cell 5, Cell 6 (or `cp` fallback), Cell 7, Cell 8, Cell 9, Cell 10, Cell 10.5, Cell 11.

In the Cell 5 log, confirm:

```
POOL_GROUPING_MODE='rating_window'
```

Run 7 — Step 5 · Check feather columns

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');
print('pool_tb_ratio_mean_snr_fwd', 'YES' if 'pool_tb_ratio_mean_snr_fwd' in df.columns else 'MISSING')"
```

Expect: `YES`.

Pool-size diagnostic (recommended for this grain):

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');
ps=df['pool_size_snr_fwd'] if 'pool_size_snr_fwd' in df.columns else None; print('pool_size median',
ps.median() if ps is not None else 'no column'); print('share pool_size<3', (ps<3).mean() if ps is not None
else 'n/a')"
```

Optional:

```
python -u talent/re_entry/army_pool_size_probe.py
```

Run 7 — Step 6 · Edit manifest panel 9

Open `talent/re_entry/manifests/army_run1_3x3_manifest.json`.

Set panel 9 `"path"` to:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
```

Save the manifest.

Run 7 — Step 7 · Run plot scripts and mosaic

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var pool_tb_ratio_mean_snr_fwd
python talent/re_entry/build_army_data_story.py
```

Run 7 — Step 8 · Verify Run 7

```
grep POOL_GROUPING_MODE pipeline_config.py
grep POOL_EXCLUDE_PEER_TB_ZERO pipeline_config.py
ls -l talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
ls -l talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3.png
```

In the basic plots log, find:

```
Overlap pool grain: POOL_GROUPING_MODE='rating_window'
```

Expect lower median pool size and/or more rows failing `pool_size ≥ 3` downstream than Run 2 or Run 6. Record panel 5 `H_sort` even if noisy — documents the sparse-bucket regime.

Run 7 — Step 9 · Archive feather and plots

Save feather:

```
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run7_ratingwindow.feather
```

Save plots:

```
bash scripts/backup_rename_suffix.sh _run7_ratingwindow --all-output
```

RUN 8 · Snapshot pool grain · straight leave-one-out · exclude zero–TB peers

What this run isolates: `active_at_snapshot` pool grain with `POOL_EXCLUDE_PEER_TB_ZERO = True` — peers with `tb_ratio == 0` dropped from pool mean and size; compare panel 5 to Run 6 (snapshot, keep peers) and to Run 3 (eval-thru, exclude peers).

Run 8 — Step 1 · Open terminal on AWS

```
cd /path/to/Network_1P_shell
conda activate TALNET39
date
```

Run 8 — Step 2 · Edit `pipeline_config.py`

```
POOL_GROUPING_MODE = 'active_at_snapshot'
POOL_EXCLUDE_PEER_TB_ZERO = True
CELL5_POOL_MEANS = True
CELL6_POOL_RANKS = True
```

Save the file.

Run 8 — Step 3 · Edit plot scripts (straight leave-one-out)

File: `talent/re_entry/army_basic_plots.py` — line 42:

```
COL_L00 = "pool_tb_ratio_mean_snr_fwd"
```

File: `talent/re_entry/army_hero_slide_plot.py` — line 44:

```
plot_var = "pool_tb_ratio_mean_snr_fwd"
```

File: `talent/re_entry/army_act2_probes.py` — line 55:

```
COL_L00_Z = "z_pool_tb_ratio_mean_snr_fwd"
```

Save all edited files.

Run 8 — Step 4 · Run notebook cells

Open `520_pipeline_cox_working.ipynb`.
Run **Cell 0**, **Cell 5**, **Cell 6** (or `cp` fallback), **Cell 7**, **Cell 8**, **Cell 9**, **Cell 10**, **Cell 10.5**, **Cell 11**.
In the Cell 5 log, confirm:

```
POOL_GROUPING_MODE='active_at_snapshot'
```

and, if printed:

```
POOL_EXCLUDE_PEER_TB_ZERO=True
```

Run 8 — Step 5 · Check feather columns

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');
print('pool_tb_ratio_mean_snr_fwd', 'YES' if 'pool_tb_ratio_mean_snr_fwd' in df.columns else 'MISSING')"
```

Expect: `YES`.
Optional pool-size diagnostic:

```
python -u talent/re_entry/army_pool_size_probe.py
```

Run 8 — Step 6 · Edit manifest panel 9

Open `talent/re_entry/manifests/army_run1_3x3_manifest.json`.
Set panel 9 `"path"` to:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
```

Save the manifest.

Run 8 — Step 7 · Run plot scripts and mosaic

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var pool_tb_ratio_mean_snr_fwd
python talent/re_entry/build_army_data_story.py
```

Run 8 — Step 8 · Verify Run 8

Panel 5 title should include `exclude tb_ratio=0 peers`.

```
grep POOL_GROUPING_MODE pipeline_config.py
grep POOL_EXCLUDE_PEER_TB_ZERO pipeline_config.py
ls -l talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
ls -l talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3.png
```

In the basic plots log, find:

```
Overlap pool grain: POOL_GROUPING_MODE='active_at_snapshot'
```

Compare panel 5 **H_sort** and panel 9 porch to Run 6 backup in `talent/re_entry/output/*/run6_activesnap/` and Run 3 backup in `talent/re_entry/output/*/run3_thru_nozero/`.

Panel 2 left histogram may still show a spike at own talent equal to zero — that is **own performance**, not the pool-exclusion toggle.

Run 8 — Step 9 · Archive feather and plots

Save feather:

```
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run8_activesnap_nozero.feather
```

Save plots:

```
bash scripts/backup_rename_suffix.sh _run8_activesnap_nozero --all-output
```

Results table — fill in for Alex

Run	Pool grain	Exclude zero-TB peers?	Peer metric on plots	H_sort (panel 5)	HERO porch note (panel 9)
1 legacy	legacy	no	straight leave-one-out		
2 eval- thru	active_at_eval_thru		straight leave-one-out		
3 eval- thru no- zero	active_at_eval_thru	yes	straight leave-one-out		
4 legacy (archived feather) minus- mean	legacy	no	minus-mean		
5 eval- thru minus- mean	active_at_eval_thru (archived feather)		minus-mean		
6 snapshot	active_at_snapshot	no	straight leave-one-out		
7 rating_ window	rating_window	no	straight leave-one-out		
8 snapshot no- zero	active_at_snapshot	yes	straight leave-one-out		

Common mistakes

Symptom	Cause	Fix
HERO error: column not in feather <code>...run1.png</code>	Passed filename to <code>--plot-var</code>	Use column name only, e.g. <code>pool_tb_ratio_mean_snr_fwd</code>
HERO error: <code>z_pool_tb_ratio_mean_snr_fwd</code> missing	Wrong column for Runs 1–3	Use <code>pool_tb_ratio_mean_snr_fwd</code>
Mosaic <code>FileNotFoundError</code> on panel 9	Manifest path ≠ HERO output filename	Edit manifest <code>"path"</code> to match exactly
Mosaic panels do not match standalone PNGs	Viewing backup subfolder PNGs	Open top-level <code>talent/re_entry/output/...</code> only
Mosaic build ran but looks old	Ran <code>bash build_army_data_story.py</code>	Use <code>python talent/re_entry/build_army_data_story.py</code>
Run 4 wrong grain	Forgot feather restore	<code>cp run1_legacy.feather → cox_analysis.feather</code>
Run 5 wrong grain	Missing or wrong archive	Save Run 2 feather in Run 2 Step 9; restore in Run 5 Step 2
Changed <code>pipeline_config.py</code> but feather unchanged	Did not re-run Cell 5–11	Re-run notebook after config change
Panel 1 shows N=39,517 but panel 2 shows n≈15,977	Different filters (cohort vs analysis sample)	Expected — not a compositor bug
Panel 2 zero spike after Run 3 exclude	Own talent zero, not peer exclusion	Expected — toggle drops peers with TB=0 only
Cell 5 <code>KeyError: 'snpsht_dt'</code> with <code>active_at_snapshot</code>	Merge duplicated <code>snpsht_dt</code> → <code>_x</code> / <code>_y</code> columns	Apply Run 6 Step 3b patch in <code>add_cum_oer_metrics_mod_working.py</code> ; Cell 0 reload

Related documents

- Pool mode detail: `talent/re_entry/rating_window_mode_explanation.md`

- Straight leave-one-out mosaic notes: [talent/re_entry/STRAIGHT_LOO_MOSAIC_AWS_GUIDE.md](#)
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Updated 2026-09-25 — Runs 6–8 full cold-start steps (no cross-run shorthand).